
GAUGE-ALIGNED ENDPOINTS AND CROSS-DIMENSIONAL TOPOLOGY FOR TEXT EMBEDDING DISTILLATION

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ABSTRACT

Embedding distillation transfers knowledge from a large teacher to a smaller student, yet their representations often differ in dimension and orientation. Existing methods handle this mismatch with learned projections, intermediate-layer supervision, or auxiliary objectives. We argue that the main challenge is making the final embedding spaces comparable. To address this, we propose GATE-KD, a final-embedding distillation framework that combines endpoint alignment with native-space topology transfer. A fixed principal-component projection first selects teacher information that fits the student dimension. A closed-form orthogonal alignment then follows the evolving student across epochs without learning extra parameters or altering the teacher structure. In parallel, a topology-based loss transfers the connectivity of examples directly between the original teacher and student spaces, despite their different dimensions. GATE-KD therefore supervises only final embeddings and adds no inference-time components. Across diverse teacher–student configurations and downstream tasks, it consistently matches or outperforms strong baselines. Further analyses show that repeated alignment improves learning even when the teacher geometry remains unchanged, while direct structure matching complements pointwise targets. Our results suggest that a carefully designed endpoint interface can replace more complex distillation pipelines.

1 INTRODUCTION

2 RELATED WORK

3 METHOD

Let $f_T : \mathcal{X} \rightarrow \mathbb{R}^{d_T}$ be a frozen teacher and f_S a student with L Transformer layers and width $d_S < d_T$. Writing $\text{norm}(\mathbf{x}) = \mathbf{x}/(\|\mathbf{x}\|_2 + \epsilon)$, we run the teacher once over an unlabeled corpus $\{x_i\}_{i=1}^N$ and cache only its final sentence embeddings, $\tilde{\mathbf{r}}_i = \text{norm}(f_T(x_i))$. Pooling and normalising the student’s final token states gives $\mathbf{z}_i \in \mathbb{S}^{d_S-1}$. The two vectors have different widths, so nothing can be compared until a map bridges them. We keep its teacher-dependent subspace fixed and update only its student-dependent orthogonal gauge.

3.1 TEACHER TARGETS

We fit the leading d_S -dimensional principal subspace $P_{\text{PCA}} \in \mathbb{R}^{d_T \times d_S}$ on the centred cache. Its orientation inside that subspace is arbitrary — invisible to every cosine-metric evaluation, but not to a point-wise loss — so we track it by orthogonal Procrustes. We select one calibration subset before training and reuse it unchanged. With $\mathbf{T}_{\text{PCA}}$ its normalised principal coordinates and $\mathbf{Z}^{(e)}$ the current student’s normalised outputs on the same sentences (Schönemann, 1966),

$$R^{(e)} = \arg \min_{R \in O(d_S)} \|\mathbf{T}_{\text{PCA}} R - \mathbf{Z}^{(e)}\|_F = UV^\top, \quad U\Sigma V^\top = \text{svd}\left(\mathbf{T}_{\text{PCA}}^\top \mathbf{Z}^{(e)}\right), \quad (1)$$

where $e = 0$ uses the initialized student and the gauge is refitted after each training epoch for the next one. The target of sentence x_i in epoch e is

$$\tau_i^{(e)} = \text{norm}\left(f_T(x_i) P_{\text{PCA}} R^{(e)}\right) \in \mathbb{S}^{d_S-1}. \quad (2)$$

The PCA factor is fitted once and frozen. Each $R^{(e)}$ is an exact closed-form update, not a learned projector. Because it is orthogonal, every refit leaves the targets’ pairwise cosines and retained variance exactly unchanged: the projection decides *what* of the teacher survives, while the rotation only tracks *where* that fixed geometry is placed relative to the evolving student.

3.2 OBJECTIVE

Only the final student embedding is supervised. The endpoint term aligns each student embedding with its fixed, dimension-matched teacher target:

$$\mathcal{L}_{\text{end}}^{(e)} = \frac{1}{B} \sum_{i=1}^B \left(1 - \langle \mathbf{z}_i, \tau_i^{(e)} \rangle\right), \quad (3)$$

where B is the minibatch size. This term fixes point-wise correspondence in the gauge selected by $P_{\text{PCA}} R^*$.

The second term matches the topology of the two embedding clouds without requiring a shared ambient dimension. Let $\text{Dgm}_k(\mathbf{X})$ denote the degree- k Vietoris–Rips persistence diagram of the row-wise normalised point cloud $\mathbf{X}$ under the chord distance $d(\mathbf{u}, \mathbf{v}) = \sqrt{2 - 2\langle \mathbf{u}, \mathbf{v} \rangle}$. For $k = 0$, the finite death times are exactly the $B - 1$ edge weights of the cloud’s minimum spanning tree. Writing their sorted values as $\delta_{(j)}^S$ and $\delta_{(j)}^T$, we use

$$\mathcal{L}_{H_0} = \frac{1}{B-1} \sum_{j=1}^{B-1} \left(\delta_{(j)}^S - \delta_{(j)}^T\right)^2, \quad (4)$$

$$\mathcal{L}_{\text{topo}} = \mathcal{L}_{H_0} + \lambda_1 W_2^2\left(\text{Dgm}_1(\mathbf{Z}), \text{Dgm}_1(\tilde{\mathbf{T}})\right), \quad (5)$$

where W_2 permits low-persistence cycles to match to the diagonal. The teacher cloud $\tilde{\mathbf{T}}$ contains the original cached embeddings $\tilde{\tau}_i \in \mathbb{R}^{d_T}$, not their projected targets. Consequently, $\mathcal{L}_{\text{topo}}$ preserves connectivity and cycles from the teacher’s own space and is independent of the endpoint interface. Setting $\lambda_1 = 0$ gives the H_0 -only variant.

The complete training objective contains only these two losses:

$$\mathcal{L}^{(e)} = \lambda_{\text{end}} \mathcal{L}_{\text{end}}^{(e)} + \lambda_{\text{topo}} \mathcal{L}_{\text{topo}}. \quad (6)$$

The endpoint term transfers sample-wise semantics, while the topological term constrains batch-level structure that a point-wise objective does not explicitly measure.

What the method does not contain. No projector is trained, no contrastive or Gram loss is added, no intermediate layer is supervised, and nothing is attached to the student at inference: the deployed model is the unmodified encoder. Training costs one teacher pass over the corpus, one truncated SVD of the cache, and one small $d_S \times d_S$ Procrustes SVD at initialization and after each non-final epoch. Student embeddings for those refits are computed only on the fixed calibration subset; ordinary training reads cached teacher embeddings and their epoch-aligned projected targets — no teacher hidden state, attention map or online teacher forward pass is used.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Teacher–student configurations. We evaluate three configurations spanning different teacher families, teacher widths, and student capacities: (a) Qwen3-Embedding-4B ($d_T = 2560$, last-token pooling) $\rightarrow$ BERT-base (109M, $d_S = 768$, 12 layers), representing the capacity-gap setting; (b) BGE-M3 ($d_T = 1024$, an XLM-R encoder with CLS pooling) $\rightarrow$ MiniLMv2-L6-H768

Table 1: Results on nine benchmarks. Avg. denotes the unweighted mean of the nine benchmark scores. Where reported, uncertainty is the sample standard deviation across three seeds.

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Avg.
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	
(a) Qwen3-Embedding-4B → BERT-base (109M)										
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	54.36
RKD	92.23±0.15	71.96±0.18	63.79±0.40	85.00±0.04	80.27±0.38	66.31±0.10	79.32±0.02	71.72±0.16	77.70±0.20	76.48±0.04
Stella	90.02±0.15	71.90±0.22	60.14±0.45	85.96±0.05	78.82±0.27	69.93±0.13	76.53±0.23	69.79±0.12	77.22±0.16	75.59±0.10
CDM	89.16±0.04	69.63±0.23	53.74±0.13	85.57±0.08	71.22±0.19	70.04±0.05	71.26±0.29	66.97±0.21	73.69±0.38	72.36±0.13
DSKD	88.87±0.07	69.05±0.09	54.02±0.56	85.90±0.05	70.02±0.31	69.63±0.21	71.07±0.10	66.52±0.17	73.52±0.01	72.07±0.03
EMO	84.96±0.84	68.93±0.65	54.80±1.52	85.05±0.04	65.75±0.22	68.80±0.13	62.00±2.71	61.75±1.85	69.54±0.37	69.07±0.89
TALAS	93.12±0.13	74.67±0.38	71.42±0.26	86.30±0.01	87.06±0.12	68.09±0.02	80.29±0.03	76.34±0.01	82.26±0.05	79.95±0.08
GATE-KD	93.47±0.12	74.18±0.25	72.50±0.47	85.45±0.08	87.24±0.12	66.57±0.08	81.85±0.06	76.40±0.07	82.52±0.08	80.02±0.03
(b) BGE-M3 → MiniLMv2-H768 (66M)										
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	56.35
RKD	91.68±0.06	74.57±0.40	62.45±0.75	85.81±0.08	79.81±0.17	63.57±0.13	76.91±0.05	69.54±0.11	75.56±0.09	75.54±0.10
Stella	88.72±0.41	73.43±0.23	61.63±0.29	86.39±0.08	74.13±0.31	66.98±0.27	73.32±0.07	68.30±0.12	72.66±0.14	73.95±0.04
CDM	87.64±0.23	70.82±0.14	56.37±0.57	85.67±0.08	67.69±0.23	66.89±0.20	70.15±0.48	66.68±0.26	69.49±0.15	71.27±0.10
DSKD	86.69±0.38	69.76±0.22	56.98±0.39	84.92±0.07	66.75±0.17	67.23±0.10	68.26±0.45	66.28±0.02	68.31±0.35	70.58±0.10
EMO	87.17±0.59	71.07±0.22	59.32±0.02	84.25±0.41	67.89±0.92	66.09±0.18	69.04±0.38	62.26±1.77	66.47±0.40	70.40±0.35
TALAS	92.44±0.10	75.58±0.15	65.13±0.21	87.09±0.09	85.39±0.14	64.39±0.12	78.14±0.06	72.56±0.13	81.03±0.05	77.97±0.03
GATE-KD	92.56±0.04	76.30±0.31	65.62±0.29	86.58±0.05	87.20±0.06	63.43±0.02	79.23±0.03	72.69±0.05	80.60±0.13	78.24±0.05
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)										
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	52.60
RKD	89.74±0.04	72.34±0.22	57.99±0.52	84.13±0.12	75.20±0.14	67.49±0.06	74.48±0.03	68.12±0.15	72.55±0.14	73.56±0.09
Stella	85.74±0.22	70.25±0.07	53.48±0.28	84.43±0.04	71.23±0.05	66.51±0.14	70.96±0.28	63.02±0.04	67.42±0.24	70.34±0.02
CDM	85.18±0.08	69.14±0.07	52.30±0.26	84.02±0.04	68.24±0.23	66.57±0.29	69.15±0.13	63.05±0.06	66.55±0.10	69.36±0.08
DSKD	85.04±0.14	69.21±0.13	52.85±0.36	84.34±0.10	67.33±0.24	65.98±0.12	69.01±0.13	63.46±0.21	66.42±0.09	69.29±0.09
EMO	85.08±0.17	68.11±0.10	53.96±0.60	84.38±0.16	65.48±0.16	65.23±0.17	68.61±0.07	60.00±0.82	65.38±0.38	68.47±0.15
TALAS	89.79±0.03	73.87±0.17	61.75±0.61	85.99±0.03	82.07±0.09	65.79±0.04	75.67±0.05	73.91±0.03	78.31±0.04	76.35±0.08
GATE-KD	91.68±0.09	74.43±0.21	66.88±0.36	85.00±0.06	83.46±0.02	67.44±0.04	77.94±0.06	72.17±0.02	78.36±0.07	77.48±0.04

(66M, $d_S = 768$, 6 layers), representing a different teacher family; and (c) Qwen3-Embedding-0.6B ($d_T = 1024$, last-token pooling) → MiniLMv2-L6-H384 (22M, $d_S = 384$, 6 layers), representing the compact-student setting.

Distillation corpus. We distill all methods on the same fixed corpus of **102,361 unlabeled sentences**, sampled once from the training splits of the nine evaluation benchmarks (SciTail; Tweet; Emotion; STS-B; Banking77; MRPC; WiC; SICK; STS12). The same corpus is used for every run in Table 1. Distillation uses only raw text: labels, pair annotations, and similarity scores are excluded from every objective. The teacher therefore provides the only supervision.

Evaluation protocol. We evaluate frozen student embeddings on nine benchmarks spanning three task families. For classification (Banking77, Emotion, Tweet), we fit a logistic-regression probe on each benchmark’s training split and report macro-F1. For pair classification (MRPC, SciTail, WiC), we map cosine similarity to $[0, 1]$ and report average precision. For semantic textual similarity (SICK, STS12, STS-B), we report the Spearman correlation with the gold scores. Avg. denotes the unweighted mean of the nine primary scores. Appendix ?? provides the split sizes and threshold protocol.

Training efficiency. Table 2 reports the observed optimization cost of the same runs. GATE-KD uses 26%, 33%, and 31% of TALAS’s measured GPU training time in configurations (a), (b), and (c), respectively, while changing Avg. by +0.07, +0.27, and +1.13 points.

Table 2: **Training efficiency on NVIDIA H200 GPUs.** Results are mean $\pm$ sample standard deviation over three seeds from the same runs as Table 1. Step time and throughput exclude the first ten warm-up steps. GPU train minutes sum all measured optimization-step times, excluding teacher-cache construction, tokenization, and final evaluation. Peak GiB is the maximum allocated memory on the busiest GPU. Bold and underlined values denote the best and second-best results within each teacher-student configuration (ties at the displayed precision are included). Jobs used both isolated and GPU-sharing schedules, so these are observed sweep costs rather than controlled single-job hardware benchmarks.

Method	Avg. $\uparrow$	ms/step $\downarrow$	samples/s $\uparrow$	GPU train min $\downarrow$	peak GiB $\downarrow$
(a) <i>Qwen3-Embedding-4B</i> $\rightarrow$ <i>BERT-base (109M)</i>					
RKD	76.48 $\pm$ 0.04	64.2 $\pm$ 0.5	1992.6 $\pm$ 15.6	4.3 $\pm$ 0.0	1.74 $\pm$ 0.02
Stella	75.59 $\pm$ 0.10	319.3 $\pm$ 24.1	402.3 $\pm$ 31.7	21.3 $\pm$ 1.6	9.30 $\pm$ 0.00
CDM	72.36 $\pm$ 0.13	469.6 $\pm$ 7.0	272.5 $\pm$ 4.0	31.4 $\pm$ 0.5	9.30 $\pm$ 0.00
DSKD	72.07 $\pm$ 0.03	255.3 $\pm$ 16.2	502.5 $\pm$ 31.0	17.1 $\pm$ 1.1	9.45 $\pm$ 0.01
EMO	69.07 $\pm$ 0.89	4581.5 $\pm$ 812.0	28.6 $\pm$ 5.6	305.1 $\pm$ 54.0	9.31 $\pm$ 0.01
TALAS	79.95 $\pm$ 0.08	174.2 $\pm$ 2.3	734.7 $\pm$ 9.9	11.6 $\pm$ 0.2	1.45 $\pm$ 0.02
GATE-KD	80.02 $\pm$ 0.03	45.5 $\pm$ 2.1	2818.4 $\pm$ 127.9	3.0 $\pm$ 0.1	<u>1.73</u> $\pm$ 0.02
(b) <i>BGE-M3</i> $\rightarrow$ <i>MiniLMv2-H768 (66M)</i>					
RKD	75.54 $\pm$ 0.10	39.0 $\pm$ 1.0	3279.7 $\pm$ 86.9	2.6 $\pm$ 0.1	<u>1.08</u> $\pm$ 0.01
Stella	73.95 $\pm$ 0.04	144.8 $\pm$ 4.6	884.2 $\pm$ 28.1	9.7 $\pm$ 0.2	2.16 $\pm$ 0.00
CDM	71.27 $\pm$ 0.10	441.2 $\pm$ 53.1	292.7 $\pm$ 33.0	29.5 $\pm$ 3.5	2.15 $\pm$ 0.01
DSKD	70.58 $\pm$ 0.10	99.3 $\pm$ 2.3	1288.5 $\pm$ 29.2	6.7 $\pm$ 0.1	2.23 $\pm$ 0.01
EMO	70.40 $\pm$ 0.35	7397.7 $\pm$ 3655.0	22.2 $\pm$ 15.1	492.5 $\pm$ 242.9	2.16 $\pm$ 0.01
TALAS	77.97 $\pm$ 0.03	97.8 $\pm$ 5.3	1310.3 $\pm$ 72.9	6.5 $\pm$ 0.4	0.86 $\pm$ 0.01
GATE-KD	78.24 $\pm$ 0.05	32.1 $\pm$ 1.5	3991.9 $\pm$ 183.7	2.2 $\pm$ 0.1	<u>1.08</u> $\pm$ 0.01
(c) <i>Qwen3-Embedding-0.6B</i> $\rightarrow$ <i>MiniLMv2-H384 (22M)</i>					
RKD	73.56 $\pm$ 0.09	39.6 $\pm$ 1.7	3234.9 $\pm$ 136.8	2.7 $\pm$ 0.1	<u>0.41</u> $\pm$ 0.00
Stella	70.34 $\pm$ 0.02	187.8 $\pm$ 50.1	722.0 $\pm$ 227.6	12.6 $\pm$ 3.4	1.40 $\pm$ 0.23
CDM	69.36 $\pm$ 0.08	542.7 $\pm$ 96.5	240.6 $\pm$ 41.1	36.2 $\pm$ 6.4	1.53 $\pm$ 0.00
DSKD	69.29 $\pm$ 0.09	129.5 $\pm$ 13.8	995.0 $\pm$ 103.2	8.7 $\pm$ 0.9	1.56 $\pm$ 0.00
EMO	68.47 $\pm$ 0.15	6531.4 $\pm$ 2136.5	21.5 $\pm$ 8.7	434.9 $\pm$ 142.2	1.53 $\pm$ 0.00
TALAS	76.35 $\pm$ 0.08	99.3 $\pm$ 1.1	1288.3 $\pm$ 14.6	6.6 $\pm$ 0.1	0.34 $\pm$ 0.00
GATE-KD	77.48 $\pm$ 0.04	30.8 $\pm$ 0.8	4160.2 $\pm$ 104.8	2.1 $\pm$ 0.1	<u>0.41</u> $\pm$ 0.00

Table 3: **Endpoint-interface ablation.** All variants optimize $\mathcal{L}_{\text{end}}$ only, with $\mathcal{L}_{H_0} = 0$. Retained energy, Gram RMSE, and k NN overlap compare the original teacher embeddings with their fixed projected targets and are therefore not applicable to jointly learned interfaces. Orthogonal Procrustes leaves all three fixed-target geometry metrics unchanged.

Configuration	Fixed-target geometry			Downstream
	Retained energy $\uparrow$	Gram RMSE $\downarrow$	k NN@10 $\uparrow$	Avg. $\uparrow$
Learned $T \rightarrow S$	N/A	N/A	N/A	56.76
Learned $S \rightarrow T$	N/A	N/A	N/A	76.29
PCA only	0.918	0.0125	0.936	76.25
+ Procrustes	0.918	0.0125	0.936	77.14
Random only	0.381	0.0345	0.729	76.09
+ Procrustes	0.381	0.0345	0.729	76.79

5 ANALYSIS

5.1 IS THE BOTTLENECK COMPRESSION OR COMPARABILITY?

Table 3 controls the endpoint objective while changing only how teacher and student coordinates are made comparable. A random subspace retains far less teacher energy than PCA yet gives a comparable downstream score. For either fixed subspace, epoch-wise Procrustes improves the result without changing the subspace itself. This suggests that endpoint comparability, rather than dimensionality reduction alone, is the operative bottleneck.

5.2 WHICH SIGNALS REQUIRE AN INTERFACE?

Table 4 separates the coordinate-dependent endpoint term from the coordinate-invariant connectivity term. Endpoint supervision provides most of the downstream improvement, while native-space H_0 supervision substantially reduces the persistent-connectivity residual. Their combination is strongest when H_0 bypasses the endpoint projection and is computed from the original teacher representation.

Table 4: **Endpoint and topology decomposition.** Rows containing the endpoint objective use the fixed PCA interface with epoch-wise Procrustes. The H_0 -only variant operates directly in the original teacher space. All results use teacher–student configuration (c) from Table 1.

Configuration	H_0 source	H_0 residual ↓	Avg. ↑
Student	—	0.3885	52.60
H_0 only	Original	0.1114	65.87
Endpoint only	—	0.1906	77.09
+ H_0	Projected	0.1268	77.11
+ H_0 (Ours)	Original	0.1182	77.48

6 CONCLUSION

REFERENCES

Peter H. Schönemann. A generalized solution of the orthogonal Procrustes problem. *Psychometrika*, 31(1):1–10, 1966.